

CSE 4412: Introduction to Quantum Computing

Homework 1 — Chapter 2 Exercises

Exercise 2.2

Statement. Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Compute $\langle +|+\rangle$, $\langle +|-\rangle$, and $\langle -|-\rangle$. What can you say about these two vectors? You may use the fact that $|0\rangle$ and $|1\rangle$ are orthonormal.

Solution. Using bilinearity of the inner product and $\langle 0|0\rangle = \langle 1|1\rangle = 1$, $\langle 0|1\rangle = \langle 1|0\rangle = 0$:

$$\begin{aligned}\langle +|+\rangle &= \frac{1}{2} (\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 + 0 + 0 + 1) = 1, \\ \langle +|-\rangle &= \frac{1}{2} (\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1 - 0 + 0 - 1) = 0, \\ \langle -|-\rangle &= \frac{1}{2} (\langle 0|0\rangle - \langle 0|1\rangle - \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 - 0 - 0 + 1) = 1.\end{aligned}$$

Since $\langle +|+\rangle = \langle -|-\rangle = 1$, both $|+\rangle$ and $|-\rangle$ are normalized. Since $\langle +|-\rangle = 0$, they are orthogonal to each other. Therefore $\{|+\rangle, |-\rangle\}$ is an orthonormal set, and since it consists of two linearly independent vectors in the 2-dimensional space $\mathbb{C}^2$, it forms an orthonormal basis (the Hadamard basis).

Exercise 2.4

Statement. Consider the following three questions:

1. Consider the $|+\rangle$ state. What is the probability of observing $|1\rangle$ after a measurement?
2. Consider $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. What is the probability of observing $|1\rangle$ after measuring that state?
3. Consider the state $|1\rangle$. What is the probability of observing $|1\rangle$ after measurement?

Solution. Recall that for a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the probability of observing outcome $|1\rangle$ is $|\beta|^2$.

1. For $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, the coefficient of $|1\rangle$ is $\beta = \frac{1}{\sqrt{2}}$, so

$$\Pr(|1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

2. For $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$, the coefficient of $|1\rangle$ is $\beta = -\frac{1}{\sqrt{2}}$, so

$$\Pr(|1\rangle) = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

3. For $|1\rangle$ itself, $\alpha = 0$ and $\beta = 1$, so

$$\Pr(|1\rangle) = |1|^2 = 1.$$

Exercise 2.5

Statement. Imagine you are given a qubit $|\psi\rangle$ but not told anything about it (i.e., you do not know its probability amplitudes). If you perform a measurement on the state and the outcome is “0,” what does this tell you about the state? Can you learn the original probability amplitudes? What if you are given as many copies of $|\psi\rangle$ as you want – what can be said then?

Solution. Write $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

A single measurement that yields outcome “0” tells you two things: (1) that $\alpha \neq 0$ (otherwise outcome “0” would have occurred with probability zero), and (2) that, due to state collapse, the state *after* the measurement is now exactly $|0\rangle$, regardless of what α and β originally were. It does *not* let you recover the original amplitudes α, β : measurement is an inherently random, information-destroying process, and a single classical outcome (one bit) cannot be inverted to reveal two continuous complex parameters. Many different states (any $\alpha \neq 0$) could have produced this same outcome, each with different probability, so a single trial is consistent with infinitely many possible original states.

If you are given many independently prepared copies of $|\psi\rangle$ (note: this must mean you are handed additional copies from the source, since the no-cloning theorem forbids copying an *unknown* qubit yourself), you can measure each copy in the computational basis and record the fraction of “0” and “1” outcomes. By the law of large numbers, as the number of copies $n \rightarrow \infty$, the empirical frequency of “0” converges to $|\alpha|^2$ (and similarly the frequency of “1” converges to $|\beta|^2$). So you can learn $|\alpha|^2$ and $|\beta|^2$ to arbitrary precision. However, even with infinitely many copies, measuring only in a single fixed basis can never reveal the relative phase between α and β (equivalently, only the magnitudes are learned, not the full complex amplitudes) – to estimate that you would additionally need to measure copies in other bases (e.g., the Hadamard basis) and combine the statistics.

Exercise 2.6

Statement. Let $p \in [0, 1]$ be a real number and consider the state

$$|\psi\rangle = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle.$$

Answer the following: (1) probability of observing $|0\rangle$ for a given p ; (2) probability of observing $|-\rangle$ for a given p ; (3)–(4) graph both probability distributions as functions of p .

Solution.

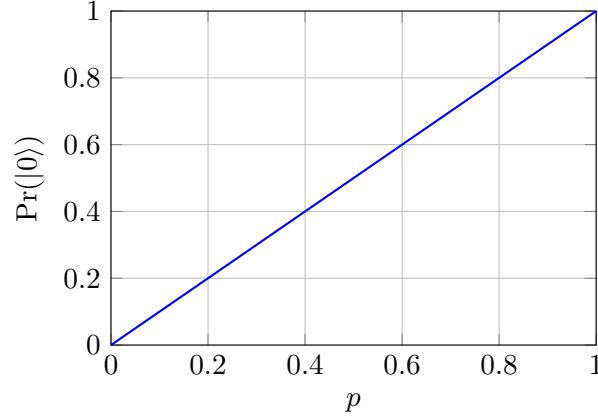
- (1) The coefficient of $|0\rangle$ is $\sqrt{p}$, so

$$\Pr(|0\rangle) = |\sqrt{p}|^2 = p.$$

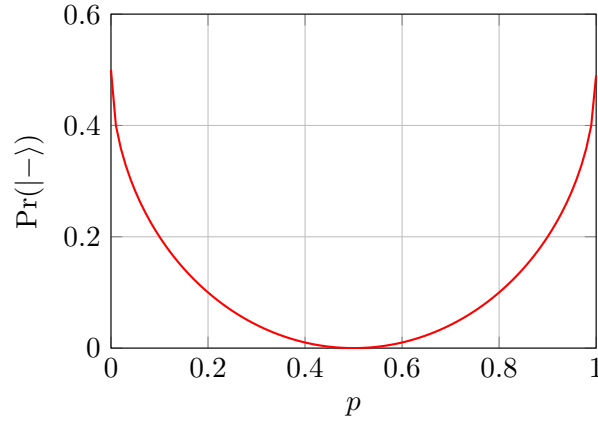
- (2) Using the inner-product rule, $\Pr(|-\rangle) = |\langle -|\psi\rangle|^2$:

$$\begin{aligned} \langle -|\psi\rangle &= \frac{1}{\sqrt{2}} (\langle 0| - \langle 1|) \cdot (\sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle) = \frac{1}{\sqrt{2}} (\sqrt{p} - \sqrt{1-p}), \\ \Pr(|-\rangle) &= \left| \frac{\sqrt{p} - \sqrt{1-p}}{\sqrt{2}} \right|^2 = \frac{1}{2} (p + (1-p) - 2\sqrt{p(1-p)}) = \frac{1}{2} - \sqrt{p(1-p)}. \end{aligned}$$

- (3) Graph of $\Pr(|0\rangle) = p$ versus p :



(4) Graph of $\text{Pr}(|-\rangle) = \frac{1}{2} - \sqrt{p(1-p)}$ versus p :



Note that $\text{Pr}(|-\rangle)$ is symmetric about $p = 1/2$, is zero exactly at $p = 1/2$ (where $|\psi\rangle = |+\rangle$, which is orthogonal to $|-\rangle$), and rises to its maximum of $1/2$ at $p = 0$ and $p = 1$ (where $|\psi\rangle$ is a computational basis state, an equal-weight superposition of $|+\rangle$ and $|-\rangle$).

Exercise 2.8

Statement. Let $|\psi\rangle$ be some d -dimensional state and let $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ be any orthonormal basis. Show that $|\psi\rangle$ may be written as

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j | \psi \rangle |b_j\rangle.$$

Solution. Since $\mathcal{B}$ is an orthonormal basis, the completeness relation tells us that

$$\sum_{j=0}^{d-1} |b_j\rangle \langle b_j| = I,$$

where I is the identity operator. Applying both sides to $|\psi\rangle$ and using the fact that $I|\psi\rangle = |\psi\rangle$:

$$|\psi\rangle = I|\psi\rangle = \left(\sum_{j=0}^{d-1} |b_j\rangle \langle b_j| \right) |\psi\rangle = \sum_{j=0}^{d-1} |b_j\rangle \langle b_j | \psi \rangle = \sum_{j=0}^{d-1} \langle b_j | \psi \rangle |b_j\rangle,$$

where the last equality holds because $\langle b_j | \psi \rangle$ is a scalar and thus commutes with the vector $|b_j\rangle$. This proves the claim. ■